<u>horizontal direction, the horizontal displacement will be lower.</u>

2 **(a)** product of mass and (linear) velocity

(b) **(i)** Since collision is elastic, total kinetic energy is the same before and after collision.

$$\cancel{\frac{1}{2}}(2\cancel{m})u^2 + \cancel{\frac{1}{2}}\cancel{m}u^2 = \cancel{\frac{1}{2}}(2\cancel{m})v_A{}^2 + \cancel{\frac{1}{2}}\cancel{m}v_B{}^2$$

$$3u^2 = 2v_A{}^2 + v_B{}^2 \quad (\text{shown})$$

(ii) **1.** Let the scattering angle of B be β respectively.
By principle of conservation of linear momentum,
Considering horizontal motion, taking right as positive,

$$2\cancel{m}(u) - \cancel{m}u = 2\cancel{m}v_A \cos\theta \quad (\because \beta = 90°)$$

$$u = 2v_A \cos\theta \quad \ldots \quad (1)$$

Considering vertical motion, taking up as positive,

$$0 = 2\cancel{m}v_A \sin\theta - \cancel{m}v_B \sin\beta$$

$$v_B = 2v_A \sin\theta \quad \ldots \quad (2)$$

$(1)^2 + (2)^2$,

$$(u)^2 + (v_B)^2 = (2v_A \cos\theta)^2 + (2v_A \sin\theta)^2$$

$$u^2 + v_B{}^2 = 4v_A{}^2 \quad \ldots \quad (3)$$

From equation in **(b)(i)**,

$$3u^2 = 2v_A{}^2 + v_B{}^2$$

$$v_A{}^2 = \frac{3u^2 - v_B{}^2}{2} \quad \ldots \quad (4)$$

Substituting (4) into (3),

$$\begin{aligned} u^2 + v_B{}^2 &= 4v_A{}^2 \\ &= 4\left(\frac{3u^2 - v_B{}^2}{2}\right) \\ &= 6u^2 - 2v_B{}^2 \\ 3v_B{}^2 &= 5u^2 \\ v_B &= \sqrt{\frac{5}{3}}u \\ &= \sqrt{\frac{5}{3}}\left(3.5 \times 10^5\right) = 4.5185 \times 10^5 = 4.5 \times 10^5 \text{ m s}^{-1} \end{aligned}$$

Alternatively,
Let the horizontal and vertical components of the velocity of particle A after collision be $v_{A,x}$ and $v_{A,y}$ respectively, and the vertical (downward) velocity of particle B after collision be v_B.
By principle of conservation of linear momentum,
Considering horizontal motion, taking right as positive,

$$2\cancel{m}(u) - \cancel{m}u = 2\cancel{m}v_{A,x} \quad \left(\because v_{B,x} = 0\right)$$

$$u = 2v_{A,x}$$

$$v_{A,x} = \frac{u}{2} \quad \ldots \quad (*)$$

Considering vertical motion, taking up as positive,

$$0 = 2\cancel{m}v_{A,y} - \cancel{m}v_B$$

$$v_B = 2v_{A,y}$$

$$v_{A,y} = \frac{v_B}{2} \quad \ldots \quad (\wedge)$$

By Pythagorean theorem,

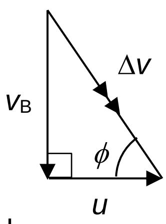
$$v_A{}^2 = v_{A,x}{}^2 + v_{A,y}{}^2$$

$$v_A{}^2 = \left(\frac{u}{2}\right)^2 + \left(\frac{v_B}{2}\right)^2 \quad \left(\text{from} (*) \text{ and } (\wedge)\right)$$

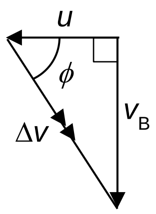
$$u^2 + v_B{}^2 = 4v_A{}^2$$

which is identical to equation (3).

(b) (ii) 2.



Alternatively,



where u, v_B and Δv are the magnitudes of the vectors.

By Pythagorean theorem,

$$\Delta v = \sqrt{u^2 + {v_B}^2}$$
$$= \sqrt{u^2 + \left(\sqrt{\frac{5}{3}}u\right)^2}$$
$$= \sqrt{\frac{8}{3}}u = \sqrt{\frac{8}{3}}\left(3.5\times10^5\right) = 5.7155\times10^5 = 5.7\times10^5 \text{ m s}^{-1}$$

$$\tan\phi = \frac{v_B}{u}$$
$$\phi = \tan^{-1}\frac{v_B}{u} = \tan^{-1}\sqrt{\frac{5}{3}} = 52°$$

52° <u>below horizontal</u>

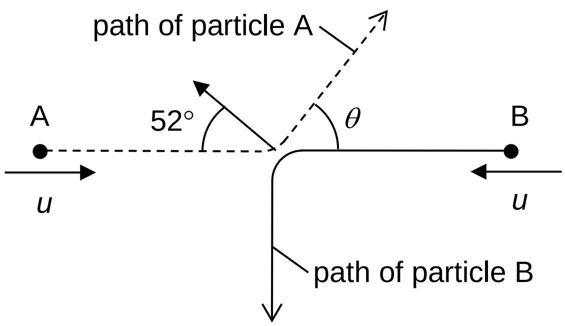
$$\Delta p = m\Delta v$$
$$= \left(1.7\times10^{-27}\right)\left(5.7155\times10^5\right)$$
$$= 9.7\times10^{-22} \text{ kg m s}^{-1}$$

change in momentum is 9.7×10^{-22} kg m s^{-1}

(b) (iii)

$$\bar{F} = \frac{\Delta p}{t}$$
$$= \frac{9.7\times10^{-22}}{1.2\times10^{-6}}$$
$$= 8.1\times10^{-16} \text{ N}$$

by Newton's third law, average force exerted by particle B on particle A is 8.1×10^{-16} N at <u>52° above horizontal</u> (in the opposite direction of change in momentum of particle B).



3 (a) 1 The resultant force acting on the body is zero.